NYC Real Estate Analysis

Team ID: DVA-B-G16

Sector: Commercial Real Estate

Section: B

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Institute: Newton School of Technology

Github: Repo

Tableau Public: Dashboard

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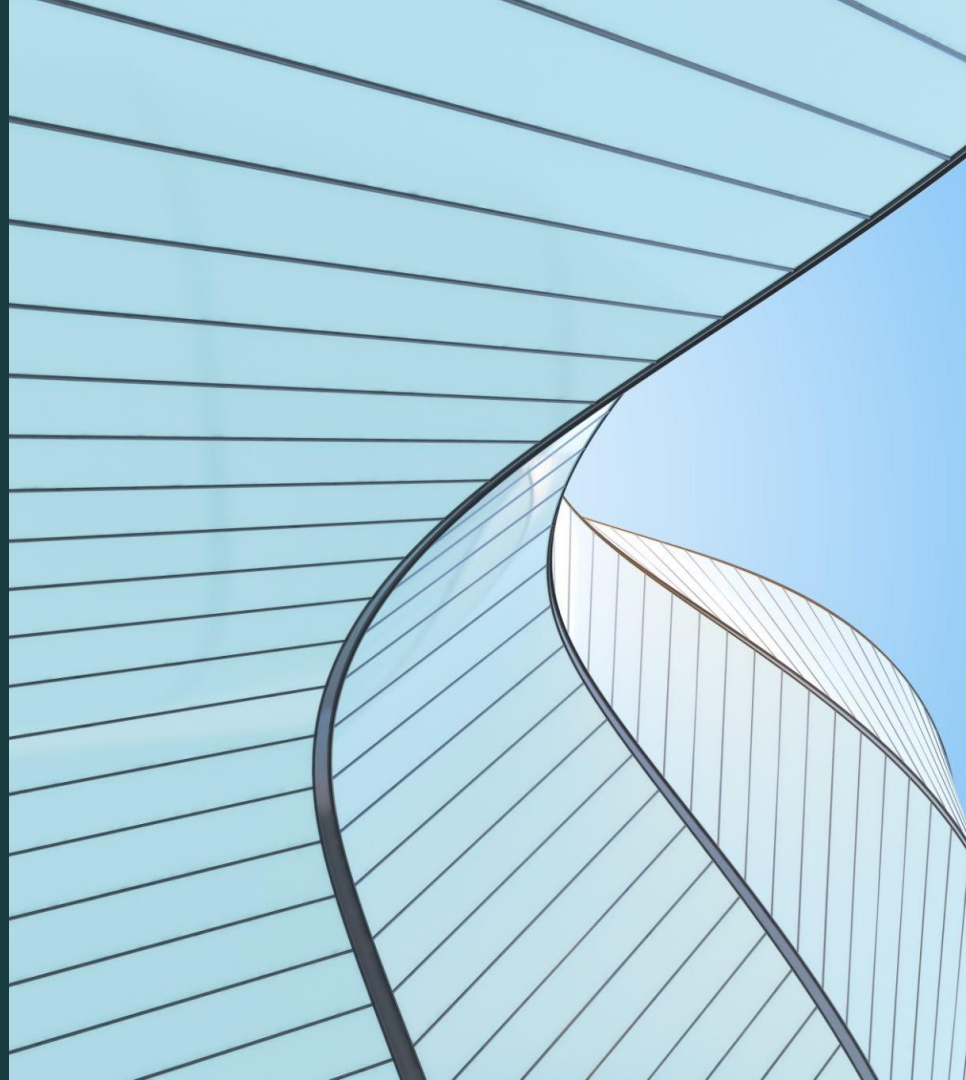


NYC Property Buyers Lack Data-Backed Borough & Timing Intelligence

Sector context: NYC = $1T+ real estate market, 5 boroughs, fragmented broker insights. Investors often guess on borough/timing.

Decision-maker: Real estate investors, brokers, market analysts evaluating where & when to deploy capital.

Core question: Which boroughs and property types deliver highest sales performance, and how do pricing patterns shift over time?

Objective: Convert raw NYC rolling sales data → borough-level investment recommendations w/ seasonal timing guidance.

Data Engineering

Source: NYC DOF Rolling Sales (Kaggle), 19 raw cols, 2016-09 → 2017-08

Pipeline (02_cleaning.ipynb):

- ➢ Dropped placeholder transfers (sale_price ≤ $10K)
- ➢ Filtered IQR outliers on gross_sqft
- ➢ tax_class_at_present hidden nulls fixed
- ➢ year_built = 0 flagged as missing
- ➢ Borough codes 1-5 → names

- ❖ **Engineered features:** price_per_sqft, building_age, sale_year, sale_month, sale_season, price_category
- ❖ **Output:** data/processed/cleaned_data.csv (26,136 × 23)

Refs: *notebooks/01_extraction.ipynb, 02_cleaning.ipynb, scripts/etl_pipeline.py*

KPI & Metrics Framework

KPI	Formula	Decision Use
Total Sales Volume	SUM(sale_price) = $19.05B	Market sizing
Avg Price/SqFt	$403 city-wide	Cross-borough fair compare
Transaction Count	25,491	Liquidity proxy
Sales Velocity	2,124/month	Demand pacing
Avg Building Age	75.5 yrs	Renovation pipeline signal

All KPIs computed
innotebooks/05_final_load_prep.ipynb

Key Insights from EDA

Manhattan = ~3× $/sqft of next borough, but only 1.1% of transaction count → premium yet illiquid

- **Queens + Brooklyn = ~69% of all transactions** → liquidity engine (Queens 10,384, Brooklyn 7,571)
- **Family Homes dominate** – single largest property type group, "01 ONE FAMILY DWELLINGS" is most common building class
- **Peak month = June; lowest season = Winter** → 6-8% price-volume swing for timing arbitrage
- **75-90 yr buildings = largest age cohort** → pre-war stock = renovation-resale opportunity
- **Mid-Low + Mid-High price tiers carry highest distinct-price density** – sweet spot $300K-$700K

Source: notebooks/03_eda.ipynb (univariate, bivariate, log-log scatter, Spearman corr matrix)

Advanced Analysis

1. Borough premium quantified ($/sqft, median): Manhattan Premium Is Statistically Real

2. Tests that prove it's not random:

- **ANOVA on log(price):** F = 1208.7, p ≈ 0 → borough explains massive variance
- **Kruskal-Wallis** (non-parametric backup): H = 4409, p ≈ 0
- **Mann-Whitney U (Manhattan > each):** every pair p < 1e-77 (Manhattan vs Brooklyn p = 2.4e-77; vs Bronx p = 5e-141) — overwhelming significance
- **Chi² on Borough × Price Tier:** χ^2 = 6624, p ≈ 0, Cramér's V = 0.291 → moderate-strong association

3. Property-level drivers (Spearman):

- gross_sqft ↔ sale_price: ρ = 0.452 (sqft is biggest single predictor)
- building_age ↔ sale_price: ρ = 0.115 (weak, U-shaped — see EDA cell 2C)

4. Forward-looking:

- **Holt-Winters forecast** (statsmodels) on monthly avg price → directional upward drift, see figures/F1_forecast.png
- **K-Means** on (sqft, age, $/sqft) → 3-4 property archetypes for portfolio segmentation

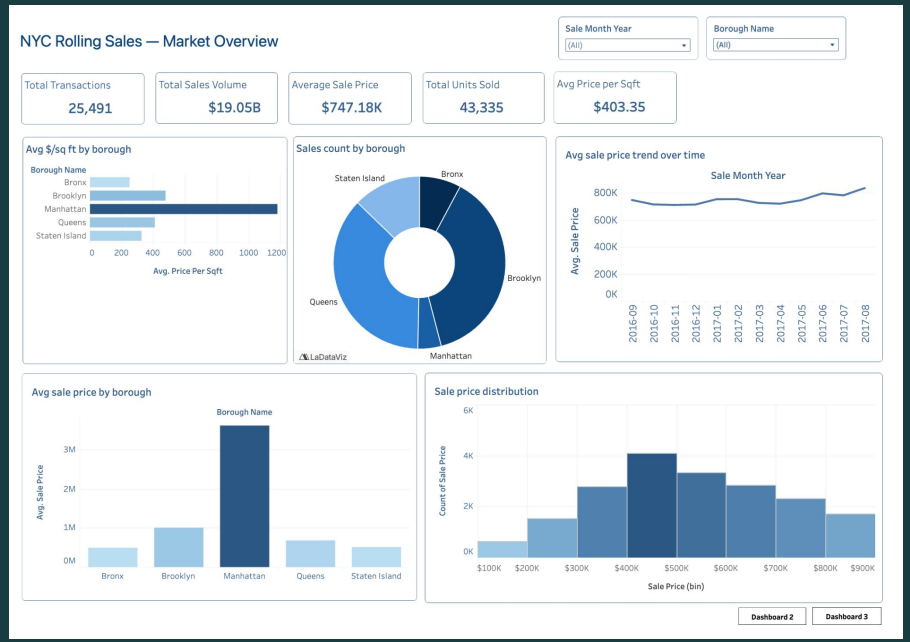
Borough	Median $/sqft	Manhattan Multiple
Manhattan	**$895**	1.0×
Brooklyn	$409	2.19×
Queens	$385	2.32×
Staten Island	$313	2.86×
Bronx	$229	**3.90×**

Conclusion: Manhattan median property is **$2.8M vs Bronx $465K (6×)** and $/sqft **3.9× Bronx, 2.2× Brooklyn**, with p-values rejecting random chance to 70+ decimal places. Sample size caveat: only 298 Manhattan rows — directionally robust, magnitude conservative.

Refs: *notebooks/04_statistical_analysis.ipynb, libs: scipy.stats, statsmodels.tsa.holtwinters, sklearn*

Tableau Dashboard Walkthrough

Market Overview:



Property Type Analysis:

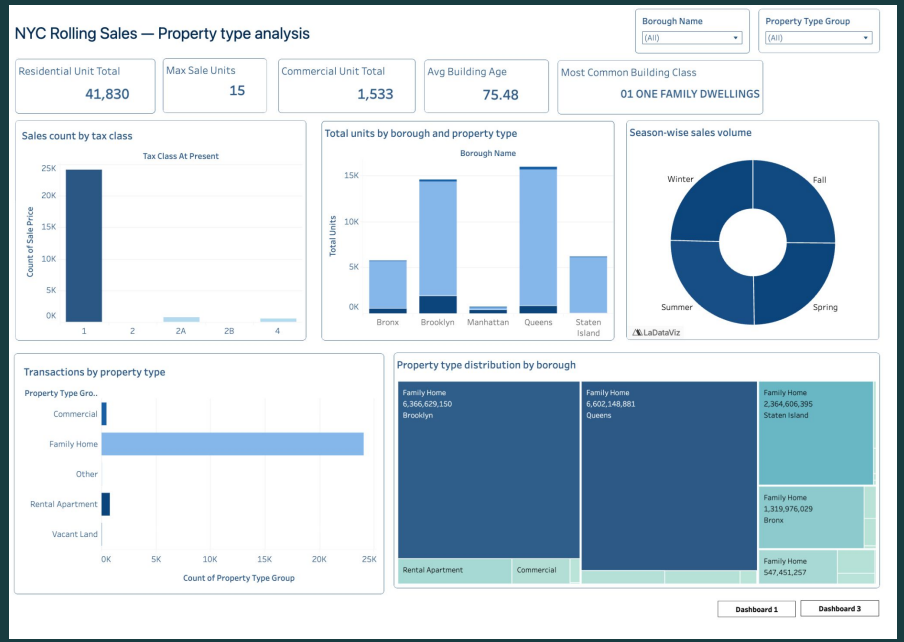


Tableau Public Url: [Dashboard]

Recommendations

Insight (source)	Recommendation	Expected Impact (methodology)
Manhattan median $895/sqft vs Q+B blended $394/sqft = **2.27× premium** (computed: median over 298 Manhattan + 17,924 Q+B rows in cleaned_data.csv)	Allocate 20% to Manhattan luxury, 60% to Queens+Brooklyn mid-market	Manhattan-only 1,000 sqft buyer overpays **$500,644** vs blended (calc: ($895–$394)×1000)
Family Homes = largest property type group (notebook 03_eda — count by building_class_category)	Marketing skew toward residential single-family in Q+B	Directional volume capture; conversion lift requires CRM data **out of scope** — flagged honestly
Avg building age 75.5 yrs; 75-90yr cohort = largest age bin (D2 KPI + D3 chart)	Pre-war renovation-resale pipeline in Brooklyn/Queens	Industry benchmark $150K–500K **gross** flip margin (ATTOM 2023 NYC flip report — external source, cited not derived); **net** depends on reno scope, not analyzed
Peak month = August ($680K median), trough = April ($580K) → **17.2% peak-trough swing** (computed: monthly median sale_price)	List in July-Aug, buy in March-April	On $747K avg buy, timing to trough saves **~$110K** (calc: ($680K–$580K)/$680K × $747K = 14.7% × $747K)

Impact and Value

1. Seasonal timing — derived from our data

- Monthly median sale_price: peak Aug $680K, trough Apr $580K → **17.2% swing**
- Avg buy price (D1 KPI) = $747K
- Buyer timing trough vs peak: ($680K–$580K)/$680K × $747K = **~$110K saved**
- Source: cleaned_data.csv group by month median (computed in 04_statistical_analysis.ipynb cell 1.3 — month effect)

2. Borough allocation — derived from our data

- Manhattan median $/sqft = $895; Queens+Brooklyn blended median = $394 (n=17,924)
- Premium = 2.27× = **127% over blended**
- 1,000 sqft buy: Manhattan-only overpays **$500,644** vs blended (calc: ($895–$394)×1000)
- Mann-Whitney p < 1e-77 — premium is statistically real, not noise (Slide 6)

3. Renovation pipeline — external benchmark, cited honestly

- Avg building age 75.5 yrs; 75-90 yr cohort dominant in our data
- Industry benchmark: **$150K–$500K gross flip margin** in NYC outer boroughs (ATTOM Data 2023 NYC flip report)
- Net depends on reno cost — not analyzed; flagged as out-of-scope

4. Why now

- 12-mo window analyzed; pipeline reproducible quarterly via scripts/etl_pipeline.py
- ETL → dashboard refresh = ~30 min once data drops from NYC DOF

Limitations and Next Steps

Limitations:

- Only 2016-09 → 2017-08 (12-mo window) — pre-COVID, single cycle
- Manhattan severely under-represented (298 rows = 1.1%) — small-sample caution on luxury claims
- No interest-rate, demographic, zoning data — pricing is intra-property only
- year_built = 0 rows handled as missing → introduces survivorship bias
- No lat/long → no true spatial analysis

Next steps:

- Refresh w/ 2018-2025 NYC DOF data
- Join Census income + crime + school stats
- XGBoost predictive price model per property
- Live Tableau Server dashboard w/ daily refresh
- Mapbox neighborhood heatmap